

Series & Convergence Tests

MATH 146 – Calculus II

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Infinite Series

Question: Given a sequence $\{a_n\}$, can you *add* the terms of the sequence together and still get a number?

That is, does $a_1 + a_2 + a_3 + \dots$ add up to a number?

Notation: We denote $a_1 + a_2 + \dots$ by $\sum_{n=1}^{\infty} a_n$ or more simply by $\sum a_n$.

Example: Infinite Series

Example: Let $a_n = 2^n$. Does $\sum_{n=0}^{\infty} a_n$ make sense?

Example: Let $a_n = \left(\frac{1}{2}\right)^n$. Does $\sum_{n=0}^{\infty} a_n$ make sense?

Infinite Series: Formal Definition

Infinite Series

Suppose $\{a_n\}$ is a sequence and let

$$S_N = a_1 + a_2 + \cdots + a_N$$

denote the N^{th} partial sum of the series. Then the series $\sum a_n$ **converges** if the sequence $\{S_N\}$ of partial sums converges.

Example: Does $\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$ converge?

Example: Infinite Series

Example (Geometric Series): Does $\sum_{n=0}^{\infty} ar^n$ converge?

Example (Telescoping Series): Does $\sum_{n=0}^{\infty} \frac{1}{k(k+1)}$ converge?

Integral Test

When do infinite series converge? It's complicated... *really complicated*.

Theorem (Integral Test)

Suppose that $f(x)$ is a **positive** function and let $a_n = f(n)$. Then

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \int_1^{\infty} f(x) dx \text{ converges}$$

Example: Integral Test

Example: Does $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge?

Example: Does $\sum_{n=1}^{\infty} \frac{1}{n}$ converge?